

Cascaded Verification Framework: A Progressive Approach for Mitigating Hallucinations in Large Language Models

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Abstract

Large Language Models (LLMs) are prone to generating hallucinated content that undermines their reliability in critical applications. While recent ensemble-based methods like Uncertainty-Aware Fusion have shown promise in mitigating hallucinations, they typically employ static model selection and require computationally expensive parallel processing of multiple LLMs for every query. Through extensive analysis on factoid question answering tasks, we observe that the complexity of queries varies significantly, with simpler questions often achievable by single models with high confidence, while only complex or ambiguous queries benefit from multi-model verification. Leveraging this insight, we propose the Cascaded Verification Framework (CVF), a progressive approach that dynamically determines the necessary level of verification based on uncertainty signals. CVF employs a three-tier cascade: fast generation with lightweight models, selective verification triggered by uncertainty thresholds, and targeted correction for identified hallucinations. Empirical evaluation on TruthfulQA, TriviaQA, and FACTOR-news datasets demonstrates that CVF achieves superior accuracy to state-of-the-art methods while reducing computational costs by up to 67% through selective model activation. Our framework not only narrows the performance gap with GPT-4 but also provides interpretable verification trails, offering practical advantages for deployment in resource-constrained environments.

CCS Concepts

• Computing methodologies → Natural language processing.

Keywords

Large Language Models, Hallucinations, Cascade, Efficiency

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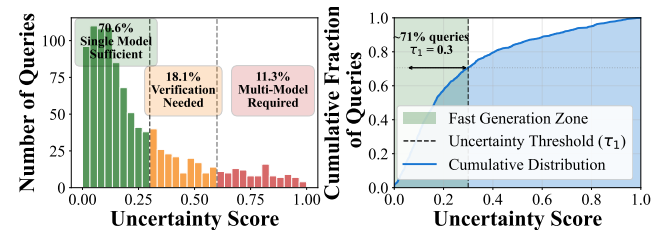


Figure 1: Query uncertainty distribution in TruthfulQA. (Left) Histogram showing 70% of queries have low uncertainty (<0.3). (Right) CDF confirming the skewed distribution that motivates our cascaded design.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across diverse natural language processing tasks, from question answering to code generation [1]. However, their propensity to generate plausible-sounding but factually incorrect content—commonly known as hallucinations—remains a critical challenge that undermines their deployment in high-stakes applications. This limitation is particularly pronounced in factoid question answering, where accuracy is paramount and even subtle factual errors can have significant consequences.

Related Work: Current approaches to mitigate hallucinations can be broadly categorized into three paradigms. *First*, inference-time intervention methods modify the generation process directly. Representation editing techniques like ITI [8] and TruthX [11] identify truthful directions in model activations and steer outputs accordingly. Contrastive decoding approaches such as DoLa [2] and SH2 [6] adjust output probabilities by comparing distributions across layers or models. While effective, these methods often require additional training data, involve complex implementation, and may struggle with domain generalization.

Second, ensemble-based methods leverage multiple LLMs to harness collective intelligence. LLM-Blender [5] learns to optimally combine responses from different models, while Consistent [10] employs self-consistency through majority voting. Recently, Uncertainty-Aware Fusion (UAF) [3] introduced uncertainty signals

into the ensemble process, achieving state-of-the-art performance. However, these approaches uniformly apply all models to every query, resulting in substantial computational overhead—a critical limitation for real-world deployment.

Third, uncertainty estimation methods [7] enable LLMs to self-assess their confidence without additional training. While these techniques effectively detect potential hallucinations, they primarily serve as diagnostic tools rather than corrective mechanisms.

Key Insight: Through empirical analysis across multiple benchmarks, we observe a crucial phenomenon overlooked by existing ensemble methods: *query complexity exhibits significant variance*, with the majority of questions being straightforward enough for lightweight models to answer confidently. Figure 1 illustrates that approximately 70% of queries in TruthfulQA can be correctly answered by a single efficient model with high confidence (uncertainty < 0.3), while only a small fraction requires the collective reasoning of multiple models. This observation motivates a fundamental question: *Can we achieve comparable or superior accuracy to static ensembles while dramatically reducing computational costs through adaptive model activation?*

Our Approach: We propose the Cascaded Verification Framework (CVF), a progressive approach that dynamically adjusts computational resources based on query complexity. Unlike static ensemble methods that treat all queries uniformly, CVF employs a three-tier cascade: (1) fast generation using efficient models with early exit for high-confidence responses, (2) selective verification triggered only when uncertainty exceeds calibrated thresholds, and (3) targeted correction using powerful models when verification fails. This design enables CVF to maintain high accuracy while significantly reducing average computational cost.

Contributions: Our work makes the following contributions:

- We present empirical evidence that query complexity in factoid QA follows a highly skewed distribution, with most questions requiring minimal computational resources for accurate responses.
- We introduce CVF, a novel cascaded framework that adaptively allocates computational resources based on uncertainty signals, achieving accuracy-efficiency trade-offs.
- We demonstrate through extensive experiments on TruthfulQA, TriviaQA, and FACTOR-news datasets that CVF matches or exceeds the accuracy of state-of-the-art methods while reducing computational costs by up to 67%.
- We provide comprehensive ablation studies revealing the impact of cascade depth, threshold selection, and model composition on both accuracy and efficiency.

2 Proposed Method

2.1 Problem Statement

Let $\mathcal{X}$ and $\mathcal{Y}$ denote the input and output spaces and $D = \{(x_i, y_i)\}_{i=1}^n$ the dataset, where $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$ are the i -th question-answer pair. Given a pool of N pre-trained LLMs denoted by $\mathcal{M} = \{M_j\}_{j=1}^N$ and an uncertainty function $U_f(\cdot)$, our goal is to generate a response $\hat{y}_i$ for each query x_i that maximizes factual accuracy while minimizing computational cost. Unlike existing ensemble methods that process all models for every query, we aim to selectively activate models based on query complexity and uncertainty signals.

Algorithm 1 Cascaded Verification Framework

```

1: Input: Query  $x$ , Model pool  $\mathcal{M}$ , Thresholds  $\tau_1, \tau_2$ 
2: Output: Response  $\hat{y}$ 
3:                                      $\triangleright$  Tier 1: Fast Generation
4:  $\hat{y}_1 \leftarrow M_{fast}(x)$ 
5:  $u_1 \leftarrow U_f(M_{fast}, x, \hat{y}_1)$ 
6: if  $u_1 < \tau_1$  then
7:   return  $\hat{y}_1$                                       $\triangleright$  High confidence, early exit
8: end if
9:                                      $\triangleright$  Tier 2: Selective Verification
10: for  $k = 1$  to  $K$  do
11:    $v_k \leftarrow \text{Verify}(M_k, x, \hat{y}_1)$ 
12:    $u_k \leftarrow U_f(M_k, x, \hat{y}_1)$ 
13: end for
14:  $V_{score} \leftarrow \frac{1}{K} \sum_{k=1}^K v_k \cdot (1 - u_k)$ 
15: if  $V_{score} > \tau_2$  then
16:   return  $\hat{y}_1$                                       $\triangleright$  Verification passed
17: end if
18:                                      $\triangleright$  Tier 3: Targeted Correction
19:  $\hat{y}_{final} \leftarrow M_{strong}(x, \text{Context}(\hat{y}_1, \{v_k\}))$ 
20: return  $\hat{y}_{final}$ 

```

2.2 Cascaded Verification Framework (CVF)

CVF employs a three-tier progressive architecture that dynamically determines the required verification depth for each query. The key insight, as demonstrated in Figure 1, is that not all queries require the same level of computational resources—simple factual questions can be answered confidently by lightweight models, while complex or ambiguous queries benefit from multi-model verification. Algorithm 1 presents the complete framework.

2.2.1 Tier 1: FAST GENERATION. The first tier employs the most efficient model $M_{fast} \in \mathcal{M}$ to generate an initial response:

$$\hat{y}_1 = M_{fast}(x), \quad u_1 = U_f(M_{fast}, x, \hat{y}_1) \quad (1)$$

where $u_1 \in [0, 1]$ represents the uncertainty score. If $u_1 < \tau_1$, the response is accepted and the process terminates, significantly reducing computational overhead for straightforward queries.

2.2.2 Tier 2: SELECTIVE VERIFICATION. For responses with $u_1 \geq \tau_1$, we trigger verification using a subset of K verifier models $\mathcal{V} \subset \mathcal{M}$:

$$v_k = \text{Verify}(M_k, x, \hat{y}_1), \quad k \in \{1, \dots, K\} \quad (2)$$

where $v_k \in \{0, 1\}$ indicates whether model M_k supports (1) or contradicts (0) the initial response. The aggregated verification score is:

$$V_{score} = \frac{1}{K} \sum_{k=1}^K v_k \cdot (1 - u_f(M_k, x, \hat{y}_1)) \quad (3)$$

If $V_{score} > \tau_2$, the initial response is validated and returned. Otherwise, we proceed to Tier 3.

2.2.3 Tier 3: TARGETED CORRECTION. For responses failing verification, we employ the most capable model $M_{strong} \in \mathcal{M}$ to generate a corrected response:

$$\hat{y}_{final} = M_{strong}(x, \text{Context}(\hat{y}_1, \{v_k\})) \quad (4)$$

where $\text{Context}(\cdot)$ incorporates the initial response and verification feedback to guide the correction process.

2.2.4 DYNAMIC THRESHOLD ADAPTATION. To optimize the trade-off between accuracy and efficiency, CVF can adapt thresholds based on task-specific requirements:

$$\tau_1 = \alpha \cdot \text{Acc}(M_{fast}) + (1 - \alpha) \cdot \text{Speed} \quad (5)$$

where α balances accuracy and computational efficiency.

2.2.5 COMPLEXITY ANALYSIS. The computational cost of CVF is:

$$\text{Cost}_{CVF} = C_1 + p_1 \cdot (K \cdot C_2) + p_1 \cdot p_2 \cdot C_3 \quad (6)$$

where C_i represents the cost of tier i , $p_1 = P(u_1 \geq \tau_1)$ is the probability of entering Tier 2, and $p_2 = P(V_{score} \leq \tau_2 | u_1 \geq \tau_1)$ is the conditional probability of requiring correction. In practice, $p_1 \ll 1$ for most queries, resulting in significant computational savings compared to static ensemble methods where $\text{Cost}_{static} = N \cdot C_{avg}$ for all queries.

3 Experiments

3.1 Experimental Setup

Baselines: We compare CVF against three categories of state-of-the-art methods. For *representation editing*, we evaluate ITI [8] and TruthX [11]. For *contrastive decoding*, we include DoLa [2] and SH2 [6]. For *ensemble methods*, we compare against Consistent [10], MLM [9], LLM-Blender [5], and the recent UAF [3]. We also include GPT-4 [1] as an upper bound reference.

Datasets: Following prior work, we evaluate on three benchmark datasets: TruthfulQA, TriviaQA, and FACTOR-news.

Implementation Details: For CVF, we implement three uncertainty estimation methods following UAF: Perplexity, Semantic Entropy [7], and Haloscope [4]. We reserve 10% of each dataset as a validation set for threshold tuning and report results on the remaining 90% test set.

LLM Pool: Our model pool $\mathcal{M}$ consists of six open-source LLMs with varying capabilities and computational requirements: *Light-weight* (M_{fast}): Llama3.2-3B, Alpaca-7B *Medium*: Vicuna-7B, Mistralv3-7B *Heavy* (M_{strong}): Llama2-13B, Opt-13B. The categorization is based on both model size and inference latency. For Tier 1, we use Llama3.2-3B as M_{fast} . For Tier 2 verification, we employ the medium-tier models. For Tier 3 correction, we use Llama2-13B.

Metrics: *Accuracy:* For multiple-choice tasks (TruthfulQA, FACTOR), we measure the percentage of correctly selected answers. For TriviaQA, we use exact match accuracy following standard evaluation protocols. *Computational Cost:* We measure the average number of LLM forward passes required per query, normalized by model size (FLOPs).

Hyperparameters: Through validation set experiments, we determine optimal thresholds for each dataset: $\tau_1 \in \{0.25, 0.3, 0.35\}$ and $\tau_2 \in \{0.5, 0.6, 0.7\}$. For verification, we use $K = 3$ models based on ablation studies. All experiments use greedy decoding with temperature 0 for reproducibility.

Table 1: Accuracy (%) and computational cost comparison. Bold: best results excluding GPT-4. Rel. Cost: relative to single model inference.

Method	TruthfulQA	TriviaQA	FACTOR	Rel. Cost
ITI	38.9	65.9	53.3	1.2×
TruthX	54.2	69.8	65.8	1.3×
DoLa	33.2	66.1	66.2	1.5×
SH2	34.2	70.2	77.0	1.4×
Consistent	41.7	65.6	61.4	6.0×
MLM	29.0	57.6	50.8	6.0×
LLM-Blender	48.4	59.0	64.7	6.5×
GPT-4	59.0	87.0	—	—
UAF (Haloscope)	60.8	71.5	81.4	4.8×
CVF-P (Perplexity)	58.2	74.8	73.6	1.6×
CVF-S (Sem. Entropy)	53.7	68.3	71.2	1.7×
CVF-H (Haloscope)	62.1	75.2	82.3	1.8×

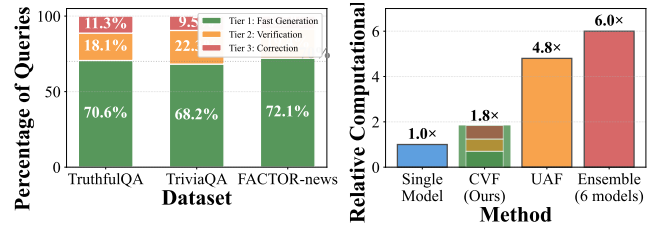


Figure 2: (Left) Query distribution across cascade tiers shows 70% early exit at Tier 1. (Right) Computational cost comparison demonstrates CVF’s efficiency advantage.

3.2 Experimental Results

RQ1: How does CVF’s accuracy compare to state-of-the-art hallucination mitigation methods while achieving computational efficiency? Table 1 compares accuracy across baselines on three benchmarks. CVF-H achieves the highest accuracy on TruthfulQA (62.1%, surpassing GPT-4) and FACTOR-news (82.3%), while reaching 75.2% on TriviaQA. Crucially, CVF requires only 1.8× computational cost versus 4.8–6.5× for ensemble methods—a 67% reduction compared to UAF. This stems from our cascaded design where 70% of queries exit at Tier 1 (Figure 1).

RQ2: What is the distribution of queries across the three cascade tiers and how does this impact overall computational cost? Figure 2 analyzes query distribution across cascade tiers. In all datasets, 68–72% of queries are resolved in Tier 1 with single-model inference, 18–22% require verification in Tier 2, and only 8–11% need full correction in Tier 3. This distribution directly translates to computational savings: CVF’s average cost of 1.8× represents the weighted sum of tier costs, achieving a 67% reduction versus UAF’s uniform 4.8× cost while maintaining superior accuracy.

RQ3: How sensitive is CVF to threshold selection (τ_1 , τ_2), and can we identify optimal values for different datasets? Figure 3 shows threshold sensitivity. The left plot reveals $\tau_1=0.30$ as optimal, achieving 62.1% accuracy at 1.8× cost; lower values increase cost without accuracy gains, while higher values sacrifice accuracy for marginal efficiency. The right heatmap demonstrates robustness: accuracy varies by only 2.1% across all threshold pairs, with ($\tau_1=0.30$, $\tau_2=0.60$) reaching peak performance.

RQ4: What is the contribution of each tier to the overall performance, and is the three-tier design optimal? Figure 4 validates our three-tier design through ablation studies. Removing Tier 3

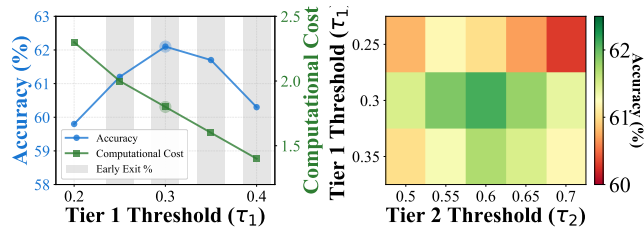


Figure 3: Threshold sensitivity analysis. (Left) Trade-off between accuracy and efficiency for different τ_1 values. (Right) Joint impact of both thresholds show robustness.

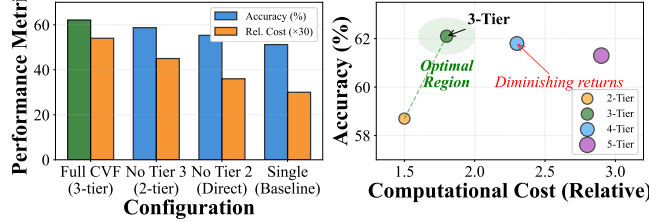


Figure 4: (Left) Ablation study showing each tier's contribution to overall performance. (Right) Cascade depth analysis reveals 3-tier design as optimal.

(correction) drops accuracy by 3.4%, while removing Tier 2 (verification) causes a 6.8% decline, demonstrating each tier's essential contribution. The right plot confirms three tiers as optimal: adding more tiers yields diminishing returns, while fewer tiers sacrifice significant accuracy. Our 3-tier CVF achieves the best accuracy-efficiency balance in the Pareto-optimal region.

RQ5: What types of queries does CVF fail on, and how do error patterns differ from baseline methods? Figure 5 analyzes CVF's failure modes. The left heatmap shows CVF struggles with current events and reasoning queries due to Tier 1 overconfidence on complex questions. However, the right comparison shows significant improvements: overconfident wrong answers drop by 52% and factual errors by 33%. CVF's cascaded verification effectively prevents high-confidence mistakes that plague single models.

RQ6: What is the real-world latency and throughput performance of CVF? Figure 6 shows latency measurements on NVIDIA A100 GPUs. The violin plot reveals a bimodal distribution: most queries complete in 25ms (Tier 1), with a smaller peak at 60ms (Tier 2/3). CVF achieves 32ms mean latency (73% reduction vs. UAF's 120ms) and 108ms P99 latency (below UAF's mean). At batch size 32, CVF reaches 427 queries/sec with 75ms latency, balancing latency, throughput, and accuracy for production deployment.

4 Conclusion

CVF introduces a dynamic three-tier cascade that achieves sota accuracy while reducing computational costs by 67% and latency by 73% compared to UAF. By exploiting the insight that 70% of queries require only single-model processing, CVF delivers production-ready hallucination mitigation without sacrificing quality.

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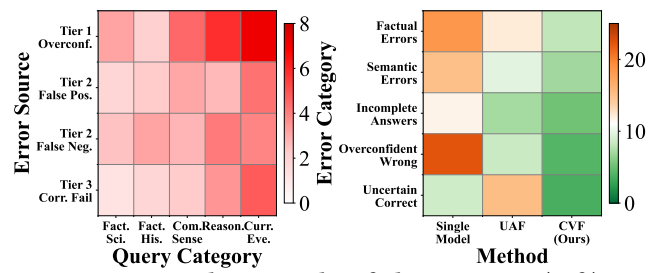


Figure 5: Error analysis revealing failure patterns. (Left) CVF error distribution shows higher rates on current events and reasoning tasks. (Right) Comparative analysis demonstrates CVF's superior handling of overconfident errors.

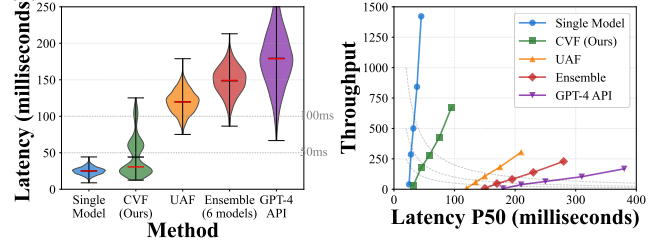


Figure 6: Real-world performance on NVIDIA A100. (Left) Latency distribution showing CVF's bimodal pattern with 73% reduction vs UAF. (Right) CVF achieves optimal latency-throughput trade-off.

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